CHAPTER 8

Prime numbers and Fermat's little theorem

Infinitude of the primes, factorisation, Fermat's little theorem

Objectives. Recognise a prime number; prove that there are infinitely many primes; factorise an integer into primes and use the factorisation (divisors, gcd); use Fermat's little theorem to compute remainders.

Prerequisites. Chapters 6 and 7; binomial coefficients; binomial formula (chapter 1).

Fluency

Fluency · No calculator, five minutes.

1. Is the integer 91 prime?
2. Give the prime numbers less than 30.
3. Factorise 360 into primes.
4. Factorise 2,026 into primes, given that 1,013 is prime.
5. Compute 2^{10} and give its remainder in the division by 11.
6. Are the integers 35 and 72 coprime?

Solution · Fluency.

1. No: $91 = 7 \times 13$.
2. 2, 3, 5, 7, 11, 13, 17, 19, 23, 29.
3. $360 = 2^3 \times 3^2 \times 5$.
4. $2{,}026 = 2 \times 1{,}013$.
5. $2^{10} = 1{,}024 = 11 \times 93 + 1$: the remainder is 1.
6. Yes: $35 = 5 \times 7$ and $72 = 2^3 \times 3^2$ have no prime factor in common.

Introductory activity

Activity 1 · A pattern in powers 35 min. For an integer $n \geq 2$ and an integer a, let $r_n(a)$ be the remainder of a^{n-1} in the division by n.

1. Compute $r_5(2)$, $r_5(3)$, $r_7(2)$, $r_7(3)$ by hand, using congruences.
2. Using the Python function `pow(a, n - 1, n)`, which returns this remainder, complete the following table.

n	5	6	7	8	9	10	11
$r_n(2)$							
$r_n(3)$							

3. For which values of n do we get 1 on both rows? What do they have in common?
4. State a conjecture. Test it for $n = 13$ and $a = 5$.
5. For $n = 7$, compute $r_7(7)$ and $r_7(14)$. How should the conjecture be made more precise?

Observing is not proving: the properties conjectured here are established in the course.

Solution · Activity.

1. $2^4 = 16 \equiv 1 \pmod 5$; $3^4 = 81 \equiv 1 \pmod 5$; $2^6 = 64 \equiv 1 \pmod 7$; $3^6 = 729 = 7 \times 104 + 1 \equiv 1 \pmod 7$.
2. Row $r_n(2)$: 1, 2, 1, 0, 4, 2, 1. Row $r_n(3)$: 1, 3, 1, 3, 0, 3, 1.
3. For $n = 5$, 7 and 11: these are the prime numbers in the table.
4. Conjecture: if p is prime, then $a^{p-1} \equiv 1 \pmod p$. For $p = 13$ and $a = 5$: $5^2 = 25 \equiv -1$, so $5^{12} = (5^2)^6 \equiv 1 \pmod{13}$.
5. $r_7(7) = r_7(14) = 0$: we must assume that a is not a multiple of p. This is Fermat's little theorem.

Teacher note · Running the activity. Insist: the converse is false. The number $341 = 11 \times 31$ is not prime, and yet $2^{340} \equiv 1 \pmod{341}$ (have it checked with `pow`). Primality tests and Carmichael numbers are "possible problems" of the syllabus.

Course notes

Prime numbers

Definition · Prime number. A natural number p is *prime* if it has exactly two positive divisors, 1 and p. Thus 1 is not prime.

Property · Smallest divisor. Every integer $n \geq 2$ has a prime divisor: its smallest divisor greater than or equal to 2. If n is not prime, this divisor p satisfies $p^2 \leq n$.

Proof. Let p be the smallest divisor of n greater than or equal to 2 (it exists, since n is one). If p were not prime, it would have a divisor d with $2 \leq d < p$, which would also divide n: a contradiction. If n is not prime, $n = pq$ with $q \geq 2$ a divisor of n, so $q \geq p$ and $n = pq \geq p^2$. □

Method · Testing whether an integer is prime. To know whether $n \geq 2$ is prime, it is enough to check that no prime p with $p^2 \leq n$ divides n. Example: 211 is divisible neither by 2, 3, 5, 7, 11, nor by 13, and $17^2 = 289 > 211$: 211 is prime.

Property · Infinitude of the primes. There are infinitely many prime numbers.

Proof. Assume there are only finitely many: $p_1, \ldots, p_k$. Let $N = p_1 p_2 \cdots p_k + 1 \geq 2$. By the previous theorem, N has a prime divisor, which is one of the p_i. This p_i divides N and the product $p_1 \cdots p_k$, so it divides their difference 1: this is impossible. The assumption is false. □

Key point · Sieve of Eratosthenes in Python. Cross out the multiples of each prime starting from its square; the numbers not crossed out are prime.

```
def crible(n):
    premier = [True] * (n + 1)
    premier[0] = premier[1] = False
    p = 2
    while p * p <= n:
        if premier[p]:
            for m in range(p * p, n + 1, p):
                premier[m] = False
        p = p + 1
    return [k for k in range(n + 1) if premier[k]]
```

There are 25 prime numbers less than 100.

Property · Prime number and product. Let p be a prime number. If $p \mid ab$, then $p \mid a$ or $p \mid b$. More generally, if p divides a product, it divides one of the factors.

Proof. If $p \nmid a$, then $\gcd(p, a) = 1$, because the only positive divisors of p are 1 and p. Gauss's theorem gives $p \mid b$. The case of a product of several factors follows by induction on their number. □

Prime factorisation

Property · Factorisation. Every integer $n \geq 2$ can be written as a product of primes, $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$, with $p_1 < p_2 < \cdots < p_k$ prime and $\alpha_i \geq 1$. This expression is unique.

Proof. *Existence*, by strong induction. The integer 2 is prime. Assume every integer in $\{2, \dots, n\}$ can be factorised. If $n+1$ is prime, we are done; otherwise $n+1 = ab$ with $2 \leq a, b \leq n$, and we put the factorisations of a and b together.

Uniqueness. If $p_1 \cdots p_r = q_1 \cdots q_s$ (repeated prime factors), p_1 divides the product of the q_j, hence one of them, which is prime: $p_1 = q_j$. We cancel and start again; the two lists of factors coincide. □

Key point · Factorising in Python. Divide by the smallest divisor $p \geq 2$ as long as possible; it is prime. When $p^2 > n$, the remaining quotient, if it exceeds 1, is prime.

```
def decomposition(n):
    facteurs = []
    p = 2
    while p * p <= n:
        while n % p == 0:
            facteurs.append(p)
            n = n // p
        p = p + 1
    if n > 1:
        facteurs.append(n)
    return facteurs
```

For example, `decomposition(360)` returns `[2, 2, 2, 3, 3, 5]`.

Property · Divisors. If $n = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$, the positive divisors of n are the integers $p_1^{\beta_1} \cdots p_k^{\beta_k}$ with $0 \leq \beta_i \leq \alpha_i$. There are $(\alpha_1+1)(\alpha_2+1)\cdots(\alpha_k+1)$ of them.

Method · GCD by factorisation. The gcd of two integers is the product of the common prime factors, each raised to the smaller of the two exponents. Example: $360 = 2^3 \times 3^2 \times 5$ and $84 = 2^2 \times 3 \times 7$ give $\gcd(360, 84) = 2^2 \times 3 = 12$.

Fermat's little theorem

Property · Fermat's little theorem. Let p be a prime number and a an integer. Then $a^p \equiv a \pmod{p}$. If moreover p does not divide a, then $a^{p-1} \equiv 1 \pmod{p}$.

Proof. *Step 1.* For $1 \leq k \leq p-1$, p divides $\binom{p}{k}$. Indeed $k\binom{p}{k} = p\binom{p-1}{k-1}$, so $p \mid k\binom{p}{k}$; since $1 \leq k < p$, p does not divide k and $\gcd(p, k) = 1$; Gauss gives $p \mid \binom{p}{k}$.

Step 2. By the binomial formula, for every integer a, $(a+1)^p = a^p + \sum_{k=1}^{p-1} \binom{p}{k} a^k + 1 \equiv a^p + 1 \pmod{p}$.

Step 3. Induction on $a \geq 0$: $0^p \equiv 0$, and if $a^p \equiv a$, then $(a+1)^p \equiv a^p + 1 \equiv a + 1 \pmod{p}$. For $a < 0$, choose an integer $b \geq 0$ such that $b \equiv a \pmod{p}$; then $a^p \equiv b^p \equiv b \equiv a$.

Step 4. If $p \nmid a$, then $p \mid a^p - a = a\left(a^{p-1} - 1\right)$ and $\gcd(p, a) = 1$; Gauss gives $p \mid a^{p-1} - 1$. □

Example 1 · A remainder. Remainder of 3^{2026} in the division by 11: 11 is prime and does not divide 3, so $3^{10} \equiv 1 \pmod{11}$. Since $2{,}026 = 10 \times 202 + 6$, $3^{2026} \equiv 3^6 = 729 \equiv 3 \pmod{11}$ (because $729 = 66 \times 11 + 3$).

Watch out. The converse is false: $2^{340} \equiv 1 \pmod{341}$, whereas $341 = 11 \times 31$.

Exercises

Exercise 8.1 · Prime or not?. Are the following integers prime? 211; 221; 1,001; 257.

Solution. 211 is prime (course). $221 = 13 \times 17$. $1{,}001 = 7 \times 11 \times 13$. For 257: none of 2, 3, 5, 7, 11, 13 divides it and $17^2 = 289 > 257$: it is prime.

Exercise 8.2 · Factorisation and divisors. Factorise 720 into primes. How many positive divisors does it have? Are there many divisors of 720 that are perfect squares?

Solution. $720 = 2^4 \times 3^2 \times 5$. Number of divisors: $(4+1)(2+1)(1+1) = 30$. A square divisor has even exponents: $2^{\beta_1}3^{\beta_2}5^{\beta_3}$ with $\beta_1 \in \{0,2,4\}$, $\beta_2 \in \{0,2\}$, $\beta_3 = 0$; there are 6 of them: 1, 4, 16, 9, 36, 144.

Beyond the syllabus · Least common multiple. The least positive common multiple of two nonzero integers, written lcm, is not part of the syllabus. It is used in the next exercise, which is not assessed.

Exercise 8.3 · GCD and LCM by factorisation. Factorise 1,260 and 1,050, then deduce their gcd and their least common multiple. Check that the product of the gcd and the lcm equals the product of the two numbers.

Solution. $1{,}260 = 2^2 \times 3^2 \times 5 \times 7$ and $1{,}050 = 2 \times 3 \times 5^2 \times 7$. GCD: $2 \times 3 \times 5 \times 7 = 210$; LCM: $2^2 \times 3^2 \times 5^2 \times 7 = 6{,}300$. And $210 \times 6{,}300 = 1{,}323{,}000 = 1{,}260 \times 1{,}050$.

Exercise 8.4 · Fermat and remainders.

1. Determine the remainder of 2^{2026} in the division by 13.
2. Prove that for every integer n, $n^{13} - n$ is divisible by 13.

Solution.

1. 13 is prime and does not divide 2: $2^{12} \equiv 1 \pmod{13}$. $2{,}026 = 12 \times 168 + 10$, so $2^{2026} \equiv 2^{10} = 1{,}024 \pmod{13}$. Now $1{,}024 = 13 \times 78 + 10$: the remainder is 10.
2. This is the first form of Fermat's little theorem, $n^{13} \equiv n \pmod{13}$.

Exercise 8.5 · Divisible by 42. Prove that for every integer n, $n^7 - n$ is divisible by 42.

Solution. $42 = 2 \times 3 \times 7$. By Fermat, $n^7 \equiv n \pmod{7}$. Modulo 2: $n^2 \equiv n$, so $n^7 \equiv n$. Modulo 3: $n^3 \equiv n$, so $n^7 = n^3 \times n^3 \times n \equiv n \times n \times n = n^3 \equiv n$. Thus 2, 3 and 7 divide $n^7 - n$; they are pairwise coprime, so (corollary of Gauss's theorem, applied twice) $42 \mid n^7 - n$.

Exercise 8.6 · Square of a prime number. Let p be a prime number greater than or equal to 5. Prove that 24 divides $p^2 - 1$.

Solution. $p^2 - 1 = (p-1)(p+1)$. Since p is odd, $p-1$ and $p+1$ are two consecutive even integers: one of them is a multiple of 4, so $8 \mid p^2 - 1$. Since p is not a multiple of 3, $p \equiv 1$ or $2 \pmod 3$, so $3 \mid p-1$ or $3 \mid p+1$. 8 and 3 are coprime: $24 \mid p^2 - 1$.

Exercise 8.7 · Mersenne numbers.

1. Using the factorisation of $z^m - 1$ (chapter 4), prove that if $n = ab$ with $a \geq 2$ and $b \geq 2$, then $2^a - 1$ divides $2^n - 1$.
2. Deduce that if $2^n - 1$ is prime, then n is prime.
3. Is the converse true? Compute $2^{11} - 1$ and check that 23 divides it.

Solution.

1. With $z = 2^a$: $2^n - 1 = z^b - 1 = (z-1)\left(z^{b-1} + \cdots + z + 1\right)$, so $2^a - 1 \mid 2^n - 1$.
2. If $n \geq 2$ is not prime, $n = ab$ as in 1, and $1 < 2^a - 1 < 2^n - 1$ is a divisor of $2^n - 1$, which is therefore not prime. Finally $2^1 - 1 = 1$ is not prime. By contraposition, if $2^n - 1$ is prime, n is prime.
3. No: 11 is prime, but $2^{11} - 1 = 2{,}047 = 23 \times 89$.

Problem

Problem · A miniature RSA cipher. Choose two primes $p = 5$ and $q = 11$, and let $n = pq = 55$ and $\varphi = (p-1)(q-1) = 40$.

1. Choose $e = 3$. Explain why e and φ are coprime, then find an integer $d \in \{1, \dots, 39\}$ such that $ed \equiv 1 \pmod{40}$.
2. Encryption: a message is an integer $m \in \{0, \dots, 54\}$; it is encrypted as the remainder c of m^e modulo n. Encrypt $m = 8$.
3. Decryption: we take for granted for now that $c^d \equiv m \pmod{n}$. Check with a calculator or Python that $17^{27} \equiv 8 \pmod{55}$.
4. We prove that $m^{ed} \equiv m \pmod{55}$ for every integer m.
   (a) Write $ed = 1 + 40k$ with k a natural number.
   (b) If $5 \nmid m$, prove that $m^{40k} \equiv 1 \pmod{5}$; deduce $m^{ed} \equiv m \pmod{5}$. What happens if $5 \mid m$?
   (c) Establish likewise $m^{ed} \equiv m \pmod{11}$, then conclude.
5. Why does the security of this cipher rest on the difficulty of factorising n when p and q are very large?

Solution · Problem.

1. $40 = 2^3 \times 5$ and 3 does not divide 40: they are coprime. $3 \times 27 = 81 = 2 \times 40 + 1$: $d = 27$.
2. $8^3 = 512 = 9 \times 55 + 17$: $c = 17$.
3. We do get 8 (for example `pow(17, 27, 55)`).
4. (a) $ed = 81 = 1 + 40 \times 2$: $k = 2$.
   (b) By Fermat, $m^4 \equiv 1 \pmod{5}$, so $m^{40k} = \left(m^4\right)^{10k} \equiv 1$ and $m^{ed} = m \times m^{40k} \equiv m \pmod{5}$. If $5 \mid m$, both sides are congruent to 0.
   (c) Modulo 11: if $11 \nmid m$, $m^{10} \equiv 1$, so $m^{40k} = \left(m^{10}\right)^{4k} \equiv 1$; if $11 \mid m$, both sides are zero modulo 11. Thus 5 and 11 divide $m^{ed} - m$ and are coprime: $55 \mid m^{ed} - m$.
5. Computing d by this method requires $\varphi = (p-1)(q-1)$, hence p and q. Knowing only n and e, factorising n would be enough to find d. For well-chosen primes with several hundred digits, no known method factorises n in a reasonable time on current computers. The security of RSA rests on this assumed difficulty: it has not been proved that every way of decrypting goes through factorising n.

Teacher note · Status of the problem. RSA is a "possible problem" of the syllabus. Have the encryption and decryption functions written in Python and tested on the 55 possible messages. Question 5 gives an idea of security; it does not call for a proof.